Who Benefits from the Minimum Wage?

A Product Market Mechanism

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Motivation

Robust empirical response to $\uparrow w_{\min}$ across countries and settings:

- Workers reallocate from low- to high-productivity firms.
  Rao & Risch (2026, *QJE*); Dustmann et al. (2022, *QJE*); Engbom & Moser (2022, *AER*); Haelbig et al. (2023, *WP*).
- High-productivity firms become **more** productive/profitable; low-productivity firms become **less**.
  Rao & Risch (2026, *QJE*); Du & Wang (2020, *Econ. Modelling*); Mayneris et al. (2018, *JDE*); Draca et al. (2011, *AEJ:Applied*); Luca & Luca (2018, *WP*).

The puzzle:

Why do firms already paying **above** the minimum wage expand and become more productive?

Outline

1. Current explanations and their limitations.
2. Our mechanism.
3. Testable predictions.

Current Explanations: Labor Market Stories

The literature attributes these patterns to:

1. **Monopsonistic labor market** (Berger et al. 2025, Econometrica; Dustmann et al. 2022, QJE; Haanwinckel 2025, WP)
   w_{min} cuts markdown of low productivity firms → they ration employment → workers reallocate up the productivity ladder
2. **Search frictions & congestion** (Dustmann et al. 2022, QJE; Engbom & Moser 2022, AER; Drechsel-Grau 2023, WP)
   Low-productivity firms cut vacancies → reduced congestion → easier recruiting for high-productivity firms
3. **Managerial slack** (Hau et al. 2020, REStud; Mayneris et al. 2018, JDE; Riley & Rosazza Bondibene 2017, Labour Econ.)
   Wage shock forces organizational improvements: investment in training, technology and reorganization.

All fundamentally **labor market** mechanisms

Problems with Labor Market Stories (1/2)

1. Monopsony & Search Frictions

Both require market frictions that are hard to sustain in low-skill labor markets:

- **Thick markets:** many similar employers, substitutable workers — the "few alternatives" premise is implausible.
- **Standardized jobs:** low specialization $\Rightarrow$ fast, cheap matching $\Rightarrow$ search frictions are mechanically small.
- **Empirically:** near-complete price pass-through (Allegretto & Reich 2018, *ILR Review*; Harasztosi & Lindner 2019, *AER*); $V \approx 1.04$ in US restaurants (Aaronson & French 2007, *JLE*).

Low-skill labor is precisely where competitive markets work best — frictions are mechanically smallest.

Problems with Labor Market Stories (2/2)

2. **Managerial Slack / Effort**
   - **Issue 1:** Why does it require a wage shock? If productivity improvements were available, profit-maximizing firms would already implement them.
   - **Issue 2:** The mechanism is **one-sided** — predicts improvements only for firms that received the shock. Our puzzle is the other side: **unaffected** high-productivity firms also expand. Draca et al. (2011, *AEJ:Applied*); Haelbig et al. (2023, *WP*)
   - **Issue 3:** w_{min} directly affects only low-wage workers, yet claimed gains are firm-wide — requiring large positive spillovers from low-skill workers to the entire organization.

Our Interpretation: Product Market Competition

Unlike labor market stories, the mechanism operates through relative prices — within the same industry:

1. **Heterogeneous cost impact:** $\Delta\psi_\omega$ larger for labor-intensive (low-productivity) firms.
2. **Aggregator shift:** The industry-wide $A(p)$ rises more than high-productivity firms' costs $\Rightarrow \downarrow \frac{\psi_\omega}{A(p)}$ for the most productive.
3. **Demand reallocation:** Lower relative prices $\rightarrow$ demand shifts to productive firms; low-productivity firms contract.
   - Generated rents masquerade as 'increase in productivity'.

→ Theoretical environment.

Why Existing Frameworks Fall Short

A small section of the literature attributes these effects to product market mechanisms (Rao & Risch 2026; Williamson 1968, *QJE*; Haucap et al. 2001, *IJIO*). These papers follow the RRC/IO tradition: Cournot competition with **homogeneous goods**.

With homogeneous goods, all firms sell at a **single market price** p^* — consumers cannot substitute toward a specific firm because its relative price fell. Demand reallocation through differential price changes is **not expressible** in this framework.

One exception: The one empirical paper implicitly requiring differentiated goods — Harasztosi & Lindner (2019, *AER*) — imposes a **proportional** cost shock: all prices rise together, and no firm's relative price changes.

Testable Predictions

If the product market mechanism operates:

- **Relative prices:** least productive firms raise prices more after $\uparrow w_{\min}$; productive firms raise prices less.
- **Market share:** productive firms gain market share within the industry; unproductive firms lose.
- **Employment:** follows output — productive firms hire more, not because of labor market pressures, but because demand shifted.

Restaurant-sector papers are the only within-industry studies, but assume homogeneous goods or putty-clay technology and draw no conclusions about productivity redistribution. **These predictions are untested.**

Conclusion

1. Minimum wage generates a **product market** effect via the price aggregator: heterogeneous cost shock $\rightarrow$ relative price shift $\rightarrow$ demand reallocation within the industry.
2. High-productivity firms gain — not through labor market channels, but because their **relative prices fall**.
3. Measured productivity gains are **competitive position effects**, not genuine efficiency improvements.
4. The right test is **within-industry**: relative prices and market shares. These predictions are currently untested.

Appendix

Model

- Regulation class & Amazon example.
- Environment and equilibrium.
- When can higher costs help a firm?
- Theorem 1: shape-changing regulation.
- CES demand with Pareto costs.
- Why measured productivity rises.

Literature

- Detailed literature summary.

The regulation class we study

Target: industry-wide rules that raise marginal costs, but whose marginal burden collapses for large/productive firms.

Amazon example:

- 2012: Acquires Kiva Systems (now Amazon Robotics, automation arm).
- 2016: Starts integrating robotics in fulfillment centers (min wage workers).
- **2018: Push for doubling the federal min wage.**
- 2022: Massive layoffs start (mainly in fulfillment centers).
- 2025: Internal leaked documents suggest that automation is going to save them 600,000 jobs over the next decade.

Environment

- Continuum of differentiated goods $\omega \in (0, 1)$, in static closed economy.
- Consumers represented by a homothetic demand system with price aggregator $A(\mathbf{p})$:

$$x_\omega = x_\omega \left(\frac{p_\omega}{A(\mathbf{p})} \right)$$

- Heterogeneous firms in monopolistic competition with marginal cost $\psi_\omega > 0$ maximize:

$$\pi_\omega = (p_\omega - \psi_\omega) x_\omega$$

Environment

Using HSA demand systems (Matsuyama & Ushchev, 2020), you can rewrite the firms maximization problem the following way:

$$\pi_\omega = (p_\omega - \psi_\omega) x_\omega \left(\frac{p_\omega}{A(\mathbf{p})} \right) = \left(1 - \frac{\psi_\omega / A(\mathbf{p})}{z_\omega} \right) Y s_\omega \left(z_\omega \right)$$

Where $s_\omega(\cdot)$ is the share function, Y is expenditure, and $A(\cdot)$ a homothetic aggregator. Finally $z_\omega \equiv \frac{p_\omega}{A(\mathbf{p})}$ is the relative price.

Key Implication: Maximizing over price levels p_ω is equivalent to maximizing over relative prices z_ω, and equilibrium profit and shares are decreasing in the **relative marginal cost** $\boldsymbol{\frac{\psi_\omega}{A(\mathbf{p})}}$: $\frac{\partial \pi_\omega^*}{\partial \psi_\omega / A(\mathbf{p})} < 0$.

When can higher costs help a firm?

From $\frac{\partial \pi^*_\omega}{\partial \psi_\omega / A(\mathbf{p})} < 0$ and $\mathbf{p}^* = \boldsymbol{\mu\psi}$, firm ω benefits from a regulatory change in marginal costs whenever its **relative marginal cost falls**:

$$\frac{\psi^R_\omega}{A(\boldsymbol{\mu\psi}^R)} < \frac{\psi_\omega}{A(\boldsymbol{\mu\psi})}$$

even if its own marginal cost rises, $\psi^R_\omega > \psi_\omega$.

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Regulations

Suppose marginal costs come from a family of continuous parametric CDFs, $\psi \sim G \in \mathcal{G}$.

Def. Regulation:

A regulation changes marginal costs to $\psi^R \sim G^R$, with $G, G^R \in \mathcal{G}$.

Why parameterize? It lets us classify regulations by the way they deform the distribution: **scale**, **location**, or **shape**.

Thm 1: Regulation that changes the shape parameter will lead to a decrease in relative cost for the most productive, and increase their market share, profit and price levels.

E.g., in Pareto costs: $G(\psi) = 1 - \left(\frac{\psi_0}{\psi}\right)^{\kappa}$, κ is the shape parameter.

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Example: CES demand with Pareto costs

The relative marginal cost for firm $\omega \in (0,1)$ would be:

$$\frac{\psi_\omega}{\mu A(\boldsymbol{\psi})} = \frac{(1-\omega)^{-\frac{1}{\boldsymbol{\kappa}}}}{\mu \left[\frac{\boldsymbol{\kappa}}{\boldsymbol{\kappa}+\sigma-1}\right]^{\frac{1}{1-\sigma}}}$$

Which implies that any decrease from any initial κ would benefit the most productive firms. If we assume that a regulation marginally decreases κ (Brazil's SPED), then every firm $\omega \in (0,\varepsilon)$ will benefit, with:

$$\varepsilon = 1 - \exp\left(-\frac{\kappa}{\kappa+\sigma-1}\right)$$

Why Measured Productivity Rises

- **Labor productivity** $\equiv$ (remuneration + **profits**) / employment.
  Riley & Rosazza Bondibene (2017). Profit is included.
- High productivity firms gain market share $\Rightarrow$ revenues rise $\Rightarrow$ profits rise $\Rightarrow$ measured productivity rises.
- **This is not** a genuine efficiency gain — it is a **competitive position** gain.
- Same logic applies to TFP: higher output at similar inputs $\Rightarrow$ positive Solow residual. Scale effects and markup changes contaminate the TFP residual.

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Detailed Literature Summary (1/2)

- **Dustmann et al. (2022, QJE):** Germany. MW $\rightarrow$ workers reallocate to high-wage firms. Attributed to monopsonistic competition. Uses cross-industry variation.
- **Engbom & Moser (2022, AER):** Brazil. High-productivity firms increase recruiting. Attributed to vacancy congestion.
- **Rao & Risch (2026, QJE):** US pass-through firms. Revenues $+4.3\%$, profits unchanged, positive entry selection. Uses Cournot framework in appendix. Cross-industry treatment definition.
- **Harasztosi & Lindner (2019, AER):** Hungary. Prices rise, employment unchanged. Market and firm-level elasticity distinction — partial product market framework. No firm heterogeneity test.

Detailed Literature Summary (2/2)

- **Aaronson & French (2007, JLE):** US restaurants. Pass-through $\approx 7\%$ per 10% MW — rules out monopsony. Representative firm model; homogeneous product.
- **Luca & Luca (2018, WP):** Yelp. Low-rated restaurants exit disproportionately. Explicit Williamson (1968) citation — intra-industry evidence consistent with our Cor. 1.
- **Riley & Rosazza Bondibene (2017, Labour Econ.):** UK. Labor productivity $\equiv$ (remuneration + profits)/employment. Productivity gains include competitive position gains.

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